

TorqSiege — Build Phase & Weapons v1

2026-09-04. Design spec for TorqWars' second event, [[2 - TorqFall (direction study v2 — standalone-event frame)]] Direction 2. Covers the two things that make TorqSiege its own experience, not just "smash a wall": (4) a build phase that becomes a **rush to build and break**, and (5) an **earn/choose weapon system** of 3D-printed / DIY car attachments. Car-durability + scale live in [[Car Fleet, Scale & Protection]]; the gating collapse-physics test lives in [[TorqSiege — Collapse Test Protocol v1]] (pointer fixed 2026-09-06 — it was never specced into Build & Fabrication Plan v1; the protocol got its own doc).

The core loop (why it's a rush)

The event runs a **never-stops clock**: Build → Siege → Reinforce → Siege → Final. Pressure comes from the clock never resetting to calm.

Phase	Time	What happens
Build	5–8 min	Teams race a countdown to build the strongest castle from a shared/earned material pile. Music + MC + shrinking clock = the rush.
Siege 1	90 s	Opposing drivers attack; score by damage/topple. Builders watch their work get wrecked.
Reinforce	2–3 min	Survivors rebuild + spend earned "reinforcement tokens" on the weak spots. Shorter clock = more panic.
Siege 2	90 s	Heavier attack (better weapons unlocked).
Final / results	—	Structure-left-standing + style points scored; reset while next teams build.

Beginner rule: building is **snap/stack only — no glue, no tools**. Pre-cut modular blocks (foam bricks, tabbed cardboard panels) so anyone builds fast. The *skill* is where you build, not how.

4. Making the build phase challenging + a rush (not just stacking)

Layered mechanics — pick a subset per event; more layers = more rush:

1. **The shrinking clock + PA countdown** — the base rush. Build music, MC counting down, a horn at zero. Anything unfinished at the horn is fair game.
2. **Central material scarcity** — one shared pile of premium blocks in the middle of the arena; teams *race to grab* the good stuff. Scarcity = hustle + light comedy chaos.
3. **Material economy (earn your bricks)** — teams start with a base kit and **earn upgrade materials** by hitting build-objectives or winning the prior siege. Reinforced blocks, wide bases, "keystone" pieces

cost tokens.

4. **Blueprint bonuses** — bonus points for building to a spec that survives: a minimum height, a gate the attacker must breach, a flag kept standing, a symmetrical wall. Rewards *smart* building, not just mass.
5. **The keystone / core objective** — every castle must protect a central "**core**" block (a balloon, a light, a mascot). Siege scoring = topple the core, not just chip the wall → gives attackers a target and builders a thing to defend. This is the single best mechanic — it turns build+siege into offense/defense with a clear win condition.
6. **Reinforcement tokens** — earned in siege 1, spent in the reinforce phase on your weak spots. Escalation + a comeback path.
7. **Sabotage window (advanced)** — a short phase where a team can spend a token to lob a "wrecking ball" (a swung foam weight) at a rival's build. Chaos, optional, showcase-only for safety.
8. **Team roles** — split a team into builders + a driver; builders hand off to their driver for the siege. Keeps everyone busy across phases (fixes the "spectator during siege" gap).

Fun-for-beginners core: you literally cannot fail to enjoy driving a car through a wall (no failure state), and building is glue-free snap-together. Layers 5 (core objective) + 1 (clock) + 2 (scarcity) give the rush without a skill wall — start there, add the rest as the format matures.

5. Weapons — earned / chosen car attachments (3D-print + DIY)

Universal strap-on mount (*amended 2026-09-06 — the fleet is RENTED, so nothing bolts, screws or glues to a car; strap-on/velcro/zip-tie only*): every siege car gets a standard front **mount plate cinched over the stock bumper with 2 velcro straps + foam padding underneath** (straps off clean in seconds at handback — the strap set is priced in [[TorqSiege — What The Event Needs]] §2). The plate carries a printed **dovetail rail** so any attachment clips on/off in seconds. Print the plate + rail once per fitted car (~8, sized to the club's standard model); attachments are swappable. Simple weapons (wedge, ram plate) can also strap direct without the rail — same rule, nothing fixed to the car. *A bolted bolt-pattern plate is a **future owned-fleet option only**.*

The attachment rack (a colourful physical wall of printed weapons = content + kid magnet):

Weapon	What it does	Print / DIY	Notes
Wedge / plow	Rides under blocks, flips them up	PLA (stiff) or plywood	The safe default; great for beginners
Battering ram	Concentrated point load to punch a weak-tab wall	PLA spike + TPU tip	Best for the "keystone" kill
Wide blade	Sweeps multiple light blocks at once	PLA / foam-core	Area damage, low precision
Grapple hook	Snag a wall section + drive away to topple	PLA hook + string	Enables the "raid/pull" mechanic — very cinematic
Forklift / lifter	Servo-driven, lift + dump a block	3D-print + micro-servo	Advanced; a hero-content weapon
Flail / wrecking arm	Dangling weighted foam ball	TPU/foam weight on chain	Showcase-only, safety-gated — never near crowd; foam-only

Filament logic: rigid parts (wedge, ram body, blade) = **PLA/PETG**; impact/safety-critical tips + the car's own armor = **TPU** (flexible, rebounds, won't shatter — see [[Car Fleet, Scale & Protection]]). Everything printable on a hobby FDM printer; DIY fallbacks (plywood wedge, coroplast blade) keep us dependency-free.

Earn / choose economy (ties weapons to the loop):

- **Draft/pick before a siege** — a "weapon wall"; each driver picks 1 of N per round (kart-pickup feel, instant, beginner-friendly). **Start here.**
- **Earn by winning** — win siege 1 → unlock a stronger weapon for siege 2 (escalation + reward).
- **Objective unlocks** — hit a build-bonus or a trick shot → earn a special weapon.
- **Loadout crafting (advanced)** — spend earned tokens to build a custom front end.

Guarantee-the-collapse note: because rubber-tired light buggies hit *soft*, weapons + weak-tab structures must be **co-designed** — the ram is only as good as the wall is pre-scored to fall. This is exactly what the [[TorqSiege — Collapse Test Protocol v1]] proves out, and what the TorqFall **Collapse Gate** round field-tests cheaply first (D1-as-R&D-for-D2).

Next: (a) spec the strap-on mount plate (velcro cinch, rented-fleet legal) + print the first 3 weapons (wedge, ram, grapple) for the collapse-test; (b) build 3 candidate wall recipes (loose-stack foam, pre-scored cardboard, weak-tab modular) and test each vs each weapon on-camera; (c) grade TorqSiege in [[Game Ideas - Graded]] *after* the test confirms it collapses well. Pairs with [[Event Ops Standards (all games)]].